

Raj Kumar Goel Institute of Technology, Ghaziabad

ST-1 (Odd Semester) 2023-24

B.TECH. II Year [Branch/Section: AIML/DS/IOT]

SEM: III

Subject Name: DSTL

Subject Code : BCS303

Max. Marks : 50

Roll No./Name : 2200331530040

Time : 1.5 Hours

Instructions :

1. Attempt the questions as per the instructions given
2. Assume missing data suitably

- | | |
|-----|--|
| CO1 | Acquire Knowledge of sets and relations for solving the problems of POSET and lattices. |
| CO2 | Apply fundamental concepts of functions and Boolean algebra for solving the problems of logical abilities. |

Section – A

Q.1	Attempt all the parts	(2 x 5 =10)		
(a)	Explain the Domain and Range of relation with example.	K4	CO1	2
(b)	Let R be the relation from the set $A = \{1,3,4\}$ on itself and defined by $R = \{(1, 1), (1, 3), (3, 3), (4, 4)\}$ then find the digraph.	K4	CO1	2
(c)	Define Modular Lattice with example.	K4	CO1	2
(d)	Draw logic diagram to represent to Boolean expression: $(XY)' + XYZ$	K2	CO2	2
(e)	Simplify the following Boolean Expression : $F = x'y + xy + xy'$	K2	CO2	2

Section – B

Q.2	Attempt all parts of the following	(4 x 5 = 20)		
(a)	Justify that for any sets A, B, and C: (i) $A - (A \cap B) = A - B$ (ii) $A - (B \cap C) = (A - B) \cup (A - C)$	K3	CO1	5
(b)	Determine whether $(D_{12}, '')$ is a Lattice or not?	K3	CO1	5
(c)	State and prove the Demorgan's Theorem.	K2	CO2	5
(d)	Prove the following Boolean Expression: (i) $A + A'B = A + B$ (ii) $(A + B).(A + C) = A + BC$	K2	CO2	5

Section – C

Q.3	Attempt any one part of the following	(10x 1 = 10)		
(a)	Show that the relation R on the set Z of integers given by $R = \{(a, b): 3 \text{ divides } a-b\}$, is an equivalence relation.	K4	CO1	10
(b)	Draw the Hasse diagram of $[P(A), \subseteq]$ (Note: ' $\subseteq$ ' stands for subset, $A = \{a, b, c\}$). Find greatest element, least element, minimal element and maximal element.	K4	CO1	10
Q.4	Attempt any one part of the following	(10x 1 = 10)		
(a)	Simplify the Boolean Expression: (i) $C(B+C)(A+B+C)$ (ii) $XY + X'Z + YZ$ (iii) $A + B(A+B) + A(A'+B)$	K2	CO2	10
(b)	Minimize following Boolean function using K-Map? $F(A,B,C,D) = \sum m(1,5,6,7,11,12,13,15)$	K2	CO2	10

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CO3 Employ the rules of propositions and predicate logic to solve the complex and logical problems.

CO4 Explore the concepts of group theory and their applications for solving the advance technological problems.

Section - A

(2 x 5 = 10)

Q.1 Attempt all the parts

(a) Define Well-formed formula with example.

K3 CO3 2

(b) Show that the statements $P \leftrightarrow Q$ and $(P \wedge Q) \vee (\sim P \wedge \sim Q)$ are equivalent.

K3 CO3 2

(c) Write the dual of $P \rightarrow Q$ and $P \leftrightarrow Q$.

K3 CO3 2

(d) If a and b are any two elements of group G then prove that $(a * b)^{-1} = b^{-1} * a^{-1}$.

K4 CO4 2

(e) Let Z be the set of integers with binary operation $*$ defined by $a * b = a + b - 2, \forall a, b \in Z$. Find the identity element of group $(Z, *)$.

K4 CO4 2

Section - B

(4 x 5 = 20)

Q.2 Attempt all parts of the following

(a) The contrapositive of a statement S is given as "if $x < 2$ then $x + 4 < 6$ ". Write the statement S and its converse and inverse.

K3 CO3 5

(b) Determine whether following proposition is Tautology or not.
 $(\bar{p} \rightarrow (q \rightarrow r)) \rightarrow ((\bar{p} \rightarrow q) \rightarrow (\bar{p} \rightarrow r))$

K3 CO3 5

(c) Prove that inverse of inverse of an element in a group is equal to element itself.

K4 CO4 5

(d) If $(\{p, q\}, *)$ is a semigroup and $p * p = q$ then show that
1. $p * q = q * p$
2. $q * q = q$

K4 CO4 5

Section - C

(10 x 1 = 10)

Q.3 Attempt any one part of the following

(a) Use rules of inference to prove the validity of the following argument.

K3 CO3 10

If Mary runs for office, She will be elected. If Mary attends the meeting, she will run for office. Either Mary will attend the meeting or she will go to India. But Mary cannot go to India. Thus "Mary will be elected"

(b) Convert the following statements in quantified expressions of predicate logic and also negate them

K3 CO3 10

1. All men are honest.
2. Some students don't live in hostel
3. All boys can run faster than all girls.
4. Not every man is perfect.

(10 x 1 = 10)

Q.4 Attempt any one part of the following

(a) Let $G = \{1, -1, i, -i\}$ with the operation of ordinary multiplication on G be an algebraic structure, where $i = \sqrt{-1}$. Determine whether G is abelian group.

K4 CO4 10

(b) Show that $(R, *)$ is an abelian group where $a * b = a + b - ab$. Consider R as set of real numbers ($R \neq 1$).

K4 CO4 10

Subject Name: DSTL Subject Code : BCS303 Total Marks : 70
 Roll No./Name: 2200331530040 Time : 3 Hours

Note Attempt all Sections. If you require any missing data, then choose suitably.

CO1	Acquire Knowledge of sets and relations for solving the problems of POSET and lattices.
CO2	Apply fundamental concepts of functions and Boolean algebra for solving the problems of logical abilities.
CO3	Employ the rules of propositions and predicate logic to solve the complex and logical problems.
CO4	Explore the concepts of group theory and their applications for solving the advance technological problems
CO5	Illustrate the principles and concepts of graph theory for solving problems related to computer science.

Section – A

Q.1	Attempt all the parts	(2 x 7 = 14)		
(a)	Let $A = \{1,2,3,4,5,6\}$ be the set and $R = \{(1,1), (1,5), (2,2), (2,3), (2,6), (3,2), (3,3), (3,6), (4,4), (5,1), (5,5), (6,2), (6,3), (6,6)\}$ be the relation defined on set A. Find Equivalence classes induced by R.	K3	CO1	2
(b)	Check whether the function $f(x) = x^2 - 1$ is injective or not for $f: R \rightarrow R$.	K1	CO2	2
(c)	Write the negation of the following statement: "If I wake up early in the morning, I will be healthy."	K3	CO3	2
(d)	Define order of an element in a group with example.	K1	CO4	2
(e)	State and justify "Every cyclic group is an abelian group".	K2	CO4	2
(f)	Define complete and regular graph.	K1	CO5	2
(g)	Explain Euler's formula. Determine number of regions if a planar graph has 30 vertices of degree 3 each.	K5	CO5	2

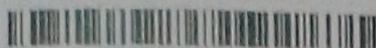
Section – B

Q.2	Attempt any three parts of the following	(7 x 3 = 21)		
(a)	Let A denote the set of real numbers and a relation R is defined on A such that $(a, b) R (c, d)$ if and only if $a^2 + b^2 = c^2 + d^2$. Justify that R is an equivalence relation.	K5	CO1	7
(b)	Solve the following Boolean functions using K-map: (i) $F(A,B,C,D) = \sum (m_0, m_1, m_2, m_4, m_5, m_6, m_8, m_9, m_{12}, m_{13}, m_{14})$ (ii) $F(A,B,C,D) = \sum (0, 2, 5, 7, 8, 10, 13, 15)$	K3	CO2	7
(c)	Show that the premises "It is not sunny this afternoon and it is colder than yesterday", "We will go swimming only if it is sunny", "If we do not go swimming, then we will take a canoe trip" and "If we take a canoe trip, then we will be home by sunset" lead to the conclusion "We will be home by sunset."	K3	CO3	7
(d)	Explain Cyclic group. Let H be a subgroup of a finite group G. Justify the statement "the order of H is a divisor of the order of G".	K5	CO4	7
(e)	Using Pigeonhole principle to find the minimum number n of integers to be selected from $S = \{1,2,3,4,5,6,7,8,9\}$ so that (i) The sum of two of the integers is even (ii) The difference of two of the n integers is 5.	K3	CO5	7

Section – C

Q.3	Attempt any one part of the following	(7x 1 = 7)		
(a)	Let L_1 be the lattice defined as $(D_6, /)$ and L_2 be the lattice $(P(S), \leq)$, where P(S) be the power set defined on set $S = \{a, b\}$. Justify that the two lattices are isomorphic.	K5	CO1	7
(b)	Define complemented lattice and then show that in a distributive lattice, if an element has a complement then this complement is unique.	K5	CO1	7

Q.4	Attempt any one part of the following	(7x 1 = 7)		
(a)	If $f: R \rightarrow R$, $g: R \rightarrow R$ and $h: R \rightarrow R$ defined by $f(x) = 3x^2 + 2$, $g(x) = 7x - 5$ and $h(x) = 1/x$. Compute the following composition functions i. $(f \circ g \circ h)(x)$ ii. $(g \circ g)(x)$ iii. $(g \circ h)(x)$	K3	CO2	7
(b)	Justify that "If $f: A \rightarrow B$ and $g: B \rightarrow C$ be one-to-one onto functions, then $g \circ f$ is also one to one onto and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ ".	K5	CO2	7
Q.5	Attempt any one part of the following	(7x 1 = 7)		
(a)	(i) Express Converse, Inverse and Contrapositive of the following statement "If $x+5 = 8$ then $x = 3$ " (ii) Show that the statements $P \leftrightarrow Q$ and $(P \wedge Q) \vee (\sim P \wedge \sim Q)$ are equivalent by using truth table.	K3	CO3	7
(b)	Define tautology, contradiction and contingency? Check whether $((p \vee q) \wedge (\sim p \vee r)) \rightarrow (q \vee r)$ is a tautology, contradiction or contingency.	K5	CO3	7
Q.6	Attempt any one part of the following	(7x 1 = 7)		
(a)	Prove that the set of residues $F = \{0, 1, 2, 3, 4\}$ modulo 5 is a field w.r.t. addition and multiplication of residue classes modulo 5. i.e. $(F, +_5, \times_5)$ is a field.	K5	CO4	7
(b)	Let $(G, *)$ and $(G', *)'$ be any two groups and let e and e' be their respective identities. If f is a homomorphism of G into G' , then prove that (i) $f(e) = e'$ (ii) $f(x^{-1}) = [f(x)]^{-1}$, $\forall x \in G$	K5	CO4	7
Q.7	Attempt any one part of the following	(7x 1 = 7)		
(a)	Explain the following (i) Euler graph and Hamiltonian graph (ii) Chromatic number of a graph (iii) Bipartite graph	K2	CO5	7
(b)	i. Justify that "In an undirected graph the total number of odd degree vertices is even". ii. Justify that "The maximum number of edges in a simple graph with n vertices is $n * (n - 1)/2$ ".	K5	CO5	7



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Roll No:

2200331530040

BTECH
(SEM III) THEORY EXAMINATION 2023-24
DISCRETE STRUCTURES & THEORY OF LOGIC

TIME: 3HRS

M.MARKS: 70

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

2 x 7 = 14

1. Attempt all questions in brief.

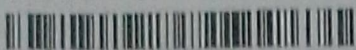
Q no.	Question	Marks	C O
a.	Determine the greatest lower bound and least upper bound of the set $\{2, 3, 6\}$, if they exist, in the Poset $(D_{24}, /)$.	2	1
b.	Express power set of each of these sets. 1) $\{\emptyset, \{\emptyset\}\}$ 2) $\{a, \{a\}\}$	2	1
c.	Investigate whether the function $f(x) = x^2 - 1$ is injective or not for $f: \mathbb{R} \rightarrow \mathbb{R}$.	2	2
d.	Express $E(x, y, z) = xy + y'z$ into its complete sum-of-products form.	2	2
e.	Construct inverse of the following statement "If I wake up early in the morning, then I will be healthy."	2	3
f.	Show that identity element is unique in a group.	2	4
g.	Compare Euler circuit and Hamiltonian circuit.	2	5

SECTION B

3 x 3 = 21

2. Attempt any three of the following:

Q no.	Question	Marks	CO
a.	Construct the Hasse Diagram for $(P(S), \subseteq)$ where $P(S)$ is a power set defined on set $S = \{a, b, c\}$. Determine whether it is a Lattice or not.	7	1
b.	Solve the following Boolean functions using K-map: (i) $F(A, B, C, D) = \sum (m_0, m_1, m_2, m_4, m_5, m_6, m_8, m_9, m_{12}, m_{13}, m_{14})$ (ii) $F(A, B, C, D) = \sum (0, 2, 5, 7, 8, 10, 13, 15)$	7	2
c.	Show the validity of the following argument: hypotheses: "It is not sunny this afternoon and it is colder than yesterday. We will go swimming only if it is sunny. If we do not go swimming, then we will take a canoe trip. If we take a canoe trip, then we will be home by sunset." conclusion: "We will be home by sunset."	7	3
d.	Let $G = \{1, -1, i, -i\}$ with the binary operation multiplication be an algebraic structure, where $i^2 = -1$ then determine whether G is an Abelian group. Also if G is cyclic Group, then determine the generator of G .	7	4
e.	Explain Pigeon hole principle. Describe generalized form of Pigeon hole principle. If 6 colors are to paint 37 homes. Show that at least 7 of them will be of same color.	7	5



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BTECH
(SEM III) THEORY EXAMINATION 2023-24
DISCRETE STRUCTURES & THEORY OF LOGIC

TIME: 3HRS

M.MARKS: 70

SECTION C**3. Attempt any one part of the following:**

7 x 1 = 7

Q no.	Question	Marks	CO
a.	Let R be a binary relation on the set of all strings of 0 and 1 such that $R = \{(a,b) : a \text{ and } b \text{ have same number of 0's}\}$. Show that whether R is reflexive, symmetric, transitive or a partial order relation.	7	1
b.	Show that $(D_{42}, /)$ is lattice. Compare the distributive and complemented lattice with example.	7	1

4. Attempt any one part of the following:

7 x 1 = 7

Q no.	Question	Marks	CO
a.	Solve the following Boolean function using K-map: $F(A,B,C) = \sum(1,2,5,7)$ and $D(0,4,6)$ using SOP.	7	2
b.	If $f: R \rightarrow R$, $g: R \rightarrow R$ and $h: R \rightarrow R$ defined by $f(x) = 3x^2 + 2$, $g(x) = 7x - 5$ and $h(x) = 1/x$. Compute the following composition functions. (i) $(f \circ g \circ h)(x)$ (ii) $(g \circ g)(x)$ (iii) $(g \circ h)(x)$	7	2

5. Attempt any one part of the following:

7 x 1 = 7

Q no.	Question	Marks	CO
a.	Test the validity of the following argument. "If there was a ball game, then traveling was difficult. If they arrived on time, then traveling was not difficult. They arrived on time. Therefore, There was no ball game."	7	3
b.	Describe $\exists$ and $\forall$ Quantifiers with example. "There is someone who got an A in the course" convert this sentence into predicate logic using quantifiers. Prove the following argument. All man are mortal. Socrates is a man. Therefore, Socrates is mortal.	7	3

6. Attempt any one part of the following:

7 x 1 = 7

Q no.	Question	Marks	CO
a.	Describe Algebraic structure, semigroup, monoid and group. Also explain the relationship among them.	7	4
b.	Consider group $G = \{1, 2, 3, 4, 5, 6\}$ under multiplication modulo 7. (a) Construct the multiplication table of G. (b) Compute 2^{-1} , 3^{-1} , 6^{-1} (c) Compute the orders and subgroups generated by 2 and 3. (d) Is G cyclic?	7	4

7. Attempt any one part of the following:

7 x 1 = 7

Q no.	Question	Marks	CO
a.	Compare bipartite and complete graph with example. Draw $K_{3,4}$ and K_5 . Explain why these two graphs are not planar.	7	5
b.	Show that $K_{3,3}$ satisfies in equality $ E \leq 3 V - 6$, but it is non-planar. (V =No. of Vertices, E =No. of Edges, R =No. of Regions)	7	5